7.5 Rotating in Three Dimensions

WORKSHEET

The question

STEP III 2022, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does $\mathbf{n} \times \mathbf{r}$ look like geometrically, when $\mathbf{r}$ is perpendicular to $\mathbf{n}$? And when $\mathbf{r}$ is parallel to $\mathbf{n}$?

STAGE 1 Crossing twice

Ask yourself Write $\mathbf{r} = \mathbf{r}_{\parallel} + \mathbf{r}_{\perp}$, where $\mathbf{r}_{\parallel} = (\mathbf{n}.\mathbf{r})\mathbf{n}$ is the component along $\mathbf{n}$. What is $f(f(\mathbf{r}))$ in terms of these?

STAGE 2 The rotation

Ask yourself Compute $g(\mathbf{n})$. Then compute $g(\mathbf{r})$ for $\mathbf{r} \perp \mathbf{n}$, using $f(f(\mathbf{r})) = -\mathbf{r}$ for such $\mathbf{r}$.

STAGE 3 The reflection

Pause and try Simplify $\mathrm{h}(\mathbf{s}) = -\mathbf{s} - 2\mathrm{f}(\mathrm{f}(\mathbf{s}))$ using part (i). What does it do to vectors along $\mathbf{n}$, and to vectors perpendicular to $\mathbf{n}$?

What's really going on

Euler's rotation theorem

Every rotation of three-dimensional space that fixes the origin is a rotation about some axis through the origin. Euler proved this in 1775. In matrix terms, every 3×3 rotation matrix has an eigenvector with eigenvalue 1, the axis. Rodrigues' formula gives the converse: for every axis $\mathbf{n}$ and angle θ, the rotation matrix is

$$\mathbf{R} = \mathbf{I} + \sin\theta\,\mathbf{K} + (1 - \cos\theta)\,\mathbf{K}^2, \qquad \mathbf{K} = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix},$$

where $\mathbf{K}$ is the matrix of $\mathbf{r} \mapsto \mathbf{n} \times \mathbf{r}$. Since $\mathbf{K}^3 = -\mathbf{K}$, the exponential series $e^{\theta\mathbf{K}} = \mathbf{I} + \theta\mathbf{K} + \frac{\theta^2}{2!}\mathbf{K}^2 + \cdots$ collapses to exactly this formula, the three-dimensional version of $e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$. This connection between rotations and exponentials of "infinitesimal rotations" $\mathbf{K}$ is the starting point of the theory of *Lie groups*.

Rotations do not commute

Hold a book flat, rotate it 90° about a vertical axis and then 90° about a horizontal axis; then start again and do the two rotations in the other order. The book ends up in different positions. Composition of rotations in three dimensions is not commutative, unlike in two dimensions. This is one reason the quaternions of Section 6.7 are useful: the rotation by θ about $\mathbf{n}$ corresponds to the quaternion $\cos\frac{\theta}{2} + \sin\frac{\theta}{2}(a\mathrm{i} + b\mathrm{j} + c\mathrm{k})$, and composing rotations is multiplying quaternions, with the non-commutativity built in. Rodrigues himself, in his 1840 paper, found the composition formula for rotations three years before Hamilton invented quaternions.

Try this next

1. **Rotations in the plane.** Try STEP III 2017, Question 2 (step.maths.org/questions/17-s3-q2). Using complex numbers, it shows that two rotations of the plane combine to give a rotation, with one exception. Find the exception, and what the combination is then.

2. **A specific rotation.** Find the matrix of the rotation by 120° about the axis in the direction $(1, 1, 1)$. Show that it permutes the coordinate axes cyclically.

3. **Powers of $\mathbf{K}$.** Show that $\mathbf{K}^2\mathbf{r} = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$ and $\mathbf{K}^3 = -\mathbf{K}$, and deduce Rodrigues' formula from the exponential series.

4. **Two reflections make a rotation.** Show that reflecting in two planes through the origin, at angle α to each other, is a rotation by 2α about their line of intersection.

Further reading

- **Simon Altmann, *Rotations, Quaternions, and Double Groups*** (Dover). The definitive account of three-dimensional rotations, including Rodrigues' contribution.
- **Simon Altmann, "Hamilton, Rodrigues, and the quaternion scandal"**, *Mathematics Magazine* 62 (1989). A short, lively history.
- **John Stillwell, *Naive Lie Theory*** (Springer). Rotations, quaternions and Lie groups for undergraduates.

The full guided solution is in Section 7.5:
https://beyondstep.keithmcnulty.org/ch07/rotations-3d.html